Executive and Strategic Vision

The CPUModel Architecture Revolution

High-Precision, Hallucination-Free, and Sustainable Artificial Intelligence

Classification: Executive Documentation (Accessible to the General Public)

Abstract

This document presents the genius behind CPUModel, an innovative technology developed by Sapiens Technology®. Today, traditional Artificial Intelligence suffers from two major problems: it consumes an absurd amount of energy (requiring highly expensive video cards) and, frequently, "hallucinates" (invents false information with great conviction). Our architecture solves this by changing the rules of the game. Using a purely logical and straightforward organization system, we eliminate heavy and confusing mathematical calculations. The goal of this document is to show, in simple terms and with everyday analogies, how we created an AI that is incredibly fast, runs on normal computers (CPUs) and, above all, **never invents facts**.

1 The Perfect Organization: Divide and Conquer

Most Artificial Intelligence models work like a giant, chaotic brain. When you ask a question, it needs to "think" about absolutely everything it knows at the same time to try and guess the answer. We followed a different and much smarter path.

1.1 The Great Library Analogy

Imagine that all the knowledge in the world is a gigantic library. Instead of throwing all the books into a single dark warehouse, we built **perfectly isolated specialist rooms**.

When you ask our AI a question, it does not scour the entire warehouse. We have an exact "map" that takes us straight to the specialist's room on the subject. This eliminates the need for complex and heavy networks. The result? A formidable saving of time and energy, as we know exactly where to look before we even start.

2 Absolute Focus: The Smart Librarian

Once inside the specialist's room, how do we find the answer without having to read all the books? In traditional AI, this requires rivers of computational processing power. In CPUModel, we created a super-sharp attention system.

2.1 The Keyword Radar

Imagine that every word in your text has a "fingerprint". Our system works like a librarian with a special radar that ignores common words (like "the", "a", "and") and focuses only on the rare words that truly carry the core meaning of your query.

Operational Highlight: The "All or Nothing" Rule
Our system acts as a flawless security filter. It only pays attention to a piece of information if it has absolute certainty that it is relevant to all parts of your question. If something is only vaguely similar, it discards it. This guarantees surgical answers.

2.2 The Treasure Hunt (Extracting What Matters)

Instead of trying to guess the context, our algorithm organizes information from the most common to the most specific. It is like panning for gold: we wash away the sand (useless information) and leave only the bright nuggets (the exact knowledge you requested).

3 The Response Engine: The Perfect Fit

The biggest breakthrough of our technology, which allows us to be so superior in efficiency, is how we connect your question to the answer. We abandoned the heavy mathematics of "matrix multiplication" (the villain that requires super-powerful video cards).

3.1 The Jigsaw Puzzle Effect

Instead of complex calculations, we transform texts into pieces of a perfect jigsaw puzzle. Our technology compares what you asked with our knowledge base using **logical yes or no operations**. Do the pieces fit together? Yes or no?

This process of checking if the pieces fit is so simple for the "brain" of a standard computer (the processor or CPU) that it does it at the speed of light, using the machine's fastest and most basic memory.

4 Creativity with Absolute Security

We want the AI to write in a fluid and natural way, not like a rigid robot. For this, we give it a slight freedom to choose words (temperature and sampling). However, the biggest fear of companies today is hallucination — when the AI invents data that does not exist.

Sapiens Trust Guarantee (End of Hallucinations):
Unlike other AIs that can "dream" with words that sound nice but are false, CPUModel has an unbreakable safety lock. Our technology guarantees that the chance of the AI using information that was not in the original documents is ***exactly ZERO****. It is only allowed to be creative using the building blocks of absolute truth. This makes our AI the definitive choice for sectors like law, healthcare, and finance, where errors are unacceptable.*

5 The End of Supercomputer Dependency

If we were to summarize the biggest victory of this project in terms of costs for your company, it would be this:

Famous language models become exponentially slower and more expensive as they read larger texts. It is as if, to read page 100 of a book, you had to read from page 1 to 99 all over again. To endure this, the market has become hostage to extremely expensive GPUs (video cards).

CPUModel breaks this curse. Since we only access the right drawer of information using our puzzle system, it does not matter if we are reading a single page or an entire encyclopedia:

- **The response speed always remains the same.**
- **The required memory does not explode.**

With this, we deliver top-tier, hyper-specialized, and secure language processing, running on normal, economical, and sustainable servers.